

SEMESTER 4

INFORMATION TECHNOLOGY

SEMESTER S4

MATHEMATICS FOR COMPUTER AND INFORMATION SCIENCE-4

(Group A)

Course Code	GAMAT401	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	Course Type	Theory

Course Objectives:

1. To provide a comprehensive understanding of fundamental concepts of graph theory including paths, cycles, trees, graph algorithms, graph coloring and matrix representations, emphasizing their applications across various disciplines.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Graphs - Basic definition, Application of graphs, finite and infinite graphs, Incidence and Degree, Isolated vertex, Pendant vertex and Null graph. Isomorphism, Sub graphs, Walks, Paths and circuits, Connected graphs, Disconnected graphs and components. [Text 1: Relevant topics from sections 1.1, 1.2, 1.3, 1.4, 1.5, 2.1, 2.2, 2.4, 2.5. Proofs of theorems 2.5, 2.7 are excluded.]	9
2	Euler graphs, Operations on Graphs, Hamiltonian paths and circuits, Travelling Salesman Problem, Connectivity, Edge connectivity, Vertex connectivity, Directed graphs, Types of directed graphs. [Text 1: Relevant topics from sections 2.6, 2.7, 2.8, 2.9, 2.10, 4.1, 4.2, 4.5, 9.1, 9.2. Proofs of theorems 4.6, 4.11, 4.12 are excluded.]	9
3	Trees- properties, Pendant vertices, Distance and centres in a tree, Rooted and binary trees, Counting trees, Spanning trees, Prim's algorithm and Kruskal's algorithm, Dijkstra's shortest path algorithm, Floyd-Warshall	9

	shortest path algorithm. [Text 1: Relevant topics from sections 3.1, 3.2, 3.3, 3.4, 3.5, 3.6, 3.7, 3.10, 11.5. Proofs of theorems 3.10, 3.16 are excluded.]	
4	Matrix representation of graphs- Adjacency matrix, Incidence Matrix, Circuit Matrix, Path Matrix, Coloring, Chromatic number, Chromatic polynomial, Greedy colouring algorithm. [Text 1: Relevant topics from sections 7.1, 7.3, 7.8, 7.9, 8.1, 8.3. Proofs of theorems 7.4, 7.7, 7.8, 8.2, 8.3, 8.5, 8.6 are excluded.]	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamental concepts of graph theory such as types of graphs, degree of a vertex, graph isomorphism, connectedness.	K2
CO2	Understand the concepts of Euler graphs, Hamiltonian graphs and connectivity.	K2
CO3	Apply Prim's and Kruskal's algorithms for finding minimum cost spanning tree and Dijkstra's and Floyd-Warshall algorithms for finding shortest paths.	K3
CO4	Illustrate various representations of graphs using matrices and apply vertex coloring in real life problems.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	-	-	-	-	-	-	-	-	2
CO2	3	3	2	-	-	-	-	-	-	-	-	2
CO3	3	3	2	2	-	-	-	-	-	-	-	2
CO4	3	3	2	2	-	-	-	-	-	-	-	2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Graph Theory with Applications to Engineering and Computer Science	Narsingh Deo	Prentice Hall India Learning Private Limited	1st edition, 1979

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Graph Theory 2e	Douglas B. West	Pearson Education India	2nd edition, 2015
2	Introduction to Graph Theory	Robin J. Wilson	Longman Group Ltd.	5th edition, 2010
3	Graph Theory with Applications	J.A. Bondy and U.S.R. Murty	Elsevier Science Publishing Co., Inc	1976

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc22_ma10/preview
2	https://onlinecourses.nptel.ac.in/noc22_ma10/preview
3	https://onlinecourses.nptel.ac.in/noc21_cs48/preview
4	https://onlinecourses.nptel.ac.in/noc21_cs48/preview

SEMESTER S4

COMPUTER NETWORKS

Course Code	PCITT402	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarise with the fundamental concepts of Computer Networks, Network hardware and Software.
2. To familiarize with the terminologies and concepts and protocols associated with the different layers of the Computer Software with the help of Reference Models.
3. To raise awareness of students with the fundamentals of Internet and internet services.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Uses of computer networks, Network hardware, Network software, Reference Models –The OSI Reference Model, TCP/IP Reference Model, Comparison of the OSI and TCP/IP Reference Models. Physical Layer: Guided Transmission Media (twisted pair, coaxial cable, Fiber Cables), Wireless Transmission (Radio Transmission, Microwave Transmission, Infrared Transmission), Communication satellites (GEO, LEO, MEO). Interfacing devices: Repeaters, Hubs, Bridges, Switches, Routers, And Gateways	8
2	Data link layer: Data link layer design issues, Error detection and correction - Error Correcting codes (Hamming codes, Reed-Solomon codes, Low-Density Parity Check codes), Error-Detecting Codes (Parity, Checksums, Cyclic Redundancy Checks), A Simplex Stop-and-Wait Protocol for a Noisy Channel, Sliding window protocols. Medium Access Control (MAC) sublayer: Channel allocation problem,	12

	Multiple access protocols (ALOHA, Slotted ALOHA, Carrier Sense Multiple Access Protocols, Collision-Free Protocols -bit map protocol, token passing, binary countdown, Wireless LAN Protocols). Ethernet: Frame format, types- Fast Ethernet, Gigabit Ethernet, 10-Gigabit Ethernet.	
3	Network Layer: Network layer design issues. Routing algorithms - The Optimality Principle, Shortest path routing, Flooding, Distance Vector Routing, Link State Routing, Multicast routing, Routing for mobile hosts. Congestion control algorithms: Approaches to Congestion Control- Choke Packets, Random Early Detection. Quality-of-service - Application Requirements, Traffic Shaping - Leaky and Token Buckets. The Network Layer in the internet: Internet Protocol, IP Version 4 Protocol, IP Addresses, IP Version 6.	12
4	Transport Layer: Transport service – Services provided to the upper layers, Transport service primitives. Socket programming, User Datagram Protocol (UDP)- User Datagram, Checksum. Real-Time Transport Protocols (RTP, RTCP), Transmission Control Protocol (TCP)- TCP Service Model, TCP Protocol, TCP Segment Header, TCP Connection Establishment, TCP Connection Release, TCP Sliding Window, TCP Congestion Control. Application Layer: Domain Name System (DNS)- Electronic mail. World Wide Web - Architectural Overview, HTTP—The Hypertext Transfer Protocol.	12

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Distinguish the terminology and concepts of computer networks and Physical Layer	K2
CO2	Illustrate different concepts of Data Link Layer including Error Control, Channel allocation and popular wired and wireless LAN technologies	K2
CO3	Describe the network layer concepts, routing algorithms, congestion control approaches and IP protocol and addressing fundamentals.	K2
CO4	Describe the features and format of different Transport layer and Application layers protocols	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	-	-	-	-	-	-	-	-	-	3
CO2	3	2	-	2	-	-	-	-	-	-	-	3
CO3	3	2	-	2	-	-	-	-	-	-	-	3
CO4	3	2	-	-	-	-	-	-	-	-	-	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Computer Networks	Andrew S. Tanenbaum	Pearson	6 th Edition 2021

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Data Communication and Networking	Behrouz A Forouzan	Tata McGraw Hill	4 th Edition 2007
2	Computer Networking - A Top-Down Approach featuring the Internet	James F Kurose	Pearson Education	7 th edition 2016
3	Computer Networks: A Systems Approach	Larry Peterson and Bruce Davie	The Morgan Kaufmann Series in Networking	5 th Edition 2011

Video Links (NPTEL, SWAYAM...)	
No.	Link ID
1	Computer Networks and Internet Protocol By Prof. Soumya Kanti Ghosh, Prof. Sandip Chakraborty https://onlinecourses.nptel.ac.in/noc22_cs19/preview

SEMESTER S4

OPERATING SYSTEMS

Course Code	PCITT403	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCITT302: Computer Organisation and Architecture	Course Type	Theory

Course Objectives:

1. To understand the mechanisms like process management in the Operating System.
2. To understand the mechanisms of process synchronization and dead lock.
3. To understand the functions of memory management, file system implementation and storage structures used in OS.
4. To understand the concepts of Virtual Machines.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Functions of OS, Types of OS - Batch, Multi programmed, Time-sharing and Real time systems, System calls, System structure - Simple structure, Layered approach, Microkernel system structure, Monolithic, Modules, Hybrid Systems, System Boot. Process Management: Process concept, Process State, PCB-Multithreading - Benefits. Case study: Linux system - process management. Process Scheduling: Basic concepts, Non Pre-emptive Scheduling, Pre-emptive Scheduling, Dispatcher, Scheduling criteria, Scheduling Algorithms - FCFS, SJF, Priority scheduling, Round Robin Scheduling.	11
2	Process Synchronization: Race Conditions - Critical Sections – Peterson’s Solution-Mutual Exclusion - Busy Waiting- Sleep and Wakeup – Bakery Algorithm-Semaphores. Deadlocks: Deadlock characteristics - conditions for deadlock - Prevention – Avoidance -Safe state, Resource Allocation Graph, Banker’s algorithm, Deadlock Detection, Recovery from dead lock.	11

3	Memory Management: Basics, Swapping, Memory Allocation - fixed partitions, variable partitions, Fragmentation, Paging, Segmentation, Virtual memory concepts, Demand paging, Page replacement algorithms - FIFO, Optimal, LRU, Allocation of frames, Thrashing.	11
4	File System: File concept - Attributes, Operations, File Types - File structure - File access methods (Sequential Access, Direct Access, Indexed Access)– File allocation methods (Contiguous, linked, and indexed allocation), Directory structure (Single-Level, Two-Level, Tree-Structured, Acyclic Graph, General Graph), Case Study: Linux file systems Disk Management: Introduction, Disk Scheduling algorithms (FCFS, SSTF, SCAN, C-SCAN, LOOK, C-LOOK) Virtual Machines: Benefits & Features- Building Blocks- Types of VMs- Virtualization & Operating System Components.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe the concepts of operating systems and apply process management and process scheduling concepts to solve problems.	K2
CO2	Illustrate process synchronization, deadlock, and deadlock – prevention and avoidance techniques.	K2
CO3	Illustrate the memory management techniques	K2
CO4	Illustrate disk scheduling algorithm to solve problems and describe the file system implementation and virtual machine.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	2	-	-	-	-	-	-	-	2
CO2	3	3	3	2	-	-	-	-	-	-	-	2
CO3	3	3	3	2	-	-	-	-	-	-	-	2
CO4	3	3	3	2	2	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Operating System Concepts	A. Silberschatz, G.Gagne and P.Galvin	Wiley Publications	9 th edition 2018

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Modern Operating Systems	Andrew S. Tanenbaum and Herbert Bos	Pearson	5 th Edition 2022
2	Operating Systems A Concept-based Approach	D M Dhamdhare	Tata McGraw Hill	2 nd Edition 2006
3	Operating Systems: Internals and Design Principles	William Stallings	Pearson	9 th Edition 2018

Video Links (NPTEL, SWAYAM...)	
No.	Link ID
1	https://archive.nptel.ac.in/courses/106/106/106106144/

SEMESTER S4

DATA SCIENCE

Course Code	PBITT404	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	UCEST105: Algorithmic Thinking with Python and PCITL307: Programming in Python Lab	Course Type	Theory

Course Objectives:

1. Students will be able to gain a comprehensive understanding of data science, including data preprocessing, exploratory data analysis, building data pipelines, implementing machine learning algorithms, and evaluating model performance.
2. Students will be able to develop practical skills in using Python and relevant data science tools to apply these concepts effectively.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introducing data science and Python: Definition of data Science, Installing latest version of python:-Step by step installation of python , Installing necessary packages of python , Introducing Jupyter. Data science process, Data loading and preprocessing with pandas- Fast and easy data loading- Data preprocessing- Data selection.	9
2	Data Pipeline, Exploratory data analysis (EDA) - Building new features- Dimensionality reduction- The detection and treatment of outliers- Validation metrics- Testing and validating- Cross-validation- Hyperparameter optimization- Feature selection- Wrapping everything in a pipeline. Preparing tools and datasets, Plotting scatter plots for relationships in data.	9
3	Machine Learning Supervised Learning: Rule-Based Classification- Using IF-THEN Rules for Classification, K-Nearest-Neighbour Classifier. Unsupervised Learning: K-means Clustering, Hierarchical Clustering, Agglomerative versus Divisive Hierarchical Clustering, Distance Measures in Algorithmic Methods, Density based Clustering - DBSCAN.	9

	Linear and Logistic Regression, Ensemble strategies.	
4	Evaluation Evaluating model performance, Confusion matrices, Precision and recall, Sensitivity and specificity, F-measure, ROC curves. Case Study: Building Your First Model in data science, Making Predictions, Evaluating the Model.	9

Suggestions on Project Topics:

No.	Project Topics
1	Churn Prediction using Machine Learning
2	Credit Analysis
3	Employee Access-Challenge as a Classification Problem
4	Fake News Detection
5	House Price Prediction using Machine Learning
6	Loan Default Prediction
7	Model Insurance Claim Severity
8	Personalized Medicine Recommending System
9	Price Recommendation using Machine Learning
10	Recommendation System for Retail Stores
11	Sales Forecasting
12	Sentiment Analysis of Product Reviews

Course Assessment Method (CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	• Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 2 sub divisions. • Each question carries 6 marks. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the fundamental concepts and scope of data science, install and utilize Python and Jupyter Notebook for data science applications.	K3
CO2	Perform exploratory data analysis (EDA), build effective data pipelines including feature engineering, dimensionality reduction, and hyperparameter optimization.	K3
CO3	Implement and evaluate various machine learning algorithms.	K3
CO4	Apply various performance metrics on Machine Learning Models.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	1	1	2	-	-	-	-	-	-	-
CO2	-	2	2	-	3	-	-	2	2	-	2	2
CO3	3	2	2	2	3	-	-	2	2	-	2	2
CO4	3	2	2	2	3	-	-	2	2	-	2	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Python Data Science Essentials	Alberto Boschetti, Luca Massaron	Packt Publishing Ltd.	3 rd Edition 2018
2	Introduction to Machine Learning with Python A Guide for Data Scientists	Andreas C. Müller and Sarah Guido	O'Reilly Media, Inc.	2017
3	Data Mining: Concepts and Techniques	Jiawei Han, Micheline Kamber, Jian Pe	Morgan Kaufmann	3 rd Edition 2012

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Data science: Concepts and practice	Vijay Kotu Bala Deshpande	Morgan Kaufmann	2nd Edition 2019
2	Data Mining Techniques	Arun K Pujari	University Press	3rd Edition 2013

Video Links (NPTEL, SWAYAM...)	
No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc20_cs80

Project Based Learning - Course Elements			
L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity		
Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S4

OBJECT ORIENTED DESIGN USING JAVA

Course Code	PEITT411	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	2:0:1:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Students should be able to understand the concepts of Software Engineering, Object oriented programming and UML
2. To provide students with a comprehensive understanding of the key concepts, activities, and management practices involved in the Requirements Elicitation, Analysis, and System Design phases of software engineering, enabling them to effectively gather, analyze, and structure requirements and design solutions that align with project goals. Work with Concurrency, Collections, and Event Handling in Java
3. To equip students with the skills to apply object-oriented design principles, including reusing pattern solutions and specifying interfaces, and to proficiently map models to code, ensuring that the software design meets functional and non-functional requirements and is implemented effectively.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Software Engineering: What is Software Engineering, Software Engineering Concepts, Software Engineering Development Activities, Managing Software Development. Introduction to Java: Objects, Classes, Constructors, Inheritance Modeling With UML: An overview of UML, Modeling Concepts, Use case Diagrams, Class Diagrams, Interaction Diagrams, State Machine Diagrams, Activity Diagrams, Diagram Organization, Diagram Extensions.	9
2	Requirements Elicitation: Introduction, Overview, Concepts, Activities, Managing Requirements Elicitation. Analysis: Introduction, Overview, Concepts, Activities, Managing Analysis	9

3	System Design: Decomposing the system - Introduction, Overview, Concepts, Activities, Addressing Design goals - Introduction, Overview, Concepts, Activities, Managing System Design. Object Design: Reusing Pattern Solutions- Introduction, Overview, Reuse Concepts, Reuse Activities, Managing Reuse.	9
4	Object Design: Specifying Interfaces- Introduction, Overview, Interface Specification Concepts, Interface Specification Activities, Managing Object Design. Mapping Models to Code: Introduction, Overview, Mapping Concepts, Mapping Activities, Managing Implementation.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Demonstrate an understanding of the fundamental concepts and activities of software engineering, including the principles of software development and management, and apply these concepts in designing object-oriented solutions using Java and UML diagrams.	K3
CO2	Illustrate the ability to effectively gather, analyze, and manage software requirements, and design system architectures by applying fundamental concepts and best practices in requirements elicitation, analysis, and system decomposition.	K3
CO3	Apply system design principles to address design goals and effectively reuse pattern solutions in object design, ensuring the development of robust, maintainable, and scalable software systems.	K3
CO4	Develop the ability to specify interfaces in object design and proficiently map models to code, ensuring accurate implementation and effective management of the software development process.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3	2	2	-	-	-	2	2	-	2
CO2	3	3	3	2	2	-	-	-	2	2	-	2
CO3	3	3	3	2	3	-	2	-	2	2	-	2
CO4	3	2	3	2	3	-	-	-	2	2	2	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Object Oriented Software Engineering using UML, Patterns and Java	Bernd Bruegge	Prentice Hall	3rd Edition 2010
2	Java: The Complete Reference	Herbert Schildt	McGraw-Hill Education	13 th Edition 2023

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Applying UML and Patterns: An Introduction to Object-oriented Analysis and Design and Iterative Development	Craig Larman	Pearson Education	3rd Edition 2005
2	Object Oriented Analysis and Design with Applications	Grady Booch	Pearson Education	3rd Edition 2014
3	Core Java Volume I - Fundamentals	Cay S. Horstmann	Prentice Hall	12th Edition 2024

Video Links (NPTEL, SWAYAM...)	
No.	Link ID
1	https://archive.nptel.ac.in/courses/106/105/106105153/

SEMESTER S4

DATA COMMUNICATION AND NETWORKING

Course Code	PEITT412	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GAEST305: Digital Electronics & Logic Design	Course Type	Theory

Course Objectives:

1. To familiarise with the fundamental concepts of analog and digital data communication.
2. To equip students to apply their knowledge in various data communication techniques for signal encoding, and error control.
3. To familiarise with the data link layer concepts and control protocols.
4. To raise awareness of students with the fundamentals of computer communication and internet services.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Data Communication Overview: Data Communications Model, Networks, The Internet. Data Transmission: Concepts and Terminology, Analog and Digital Data Transmission, Transmission Impairments, Channel capacity. Transmission Media: Guided Transmission Media, Wireless Transmission, Wireless Propagation.	6
2	Signal Encoding Techniques: Digital Data to Digital Signals (NRZ-L, NRZ-I, Bipolar- AMI, Pseudo ternary, Manchester, Differential Manchester, Scrambling techniques), Digital Data to Analog Signals (ASK, PSK, FSK), Analog Data to Digital Signals (PCM, DM), Analog Data to Analog Signals. Digital Data Communication Techniques: Asynchronous and Synchronous Transmission, Types of Errors, Error Detection (parity check, CRC), Error Correction, Line Configurations.	11

3	Data Link Control Protocols: Flow Control, Error Control, Multiplexing - Frequency-Division Multiplexing, Synchronous Time-Division Multiplexing, Statistical Time-Division Multiplexing, Spread Spectrum - The Concept of Spread Spectrum - Frequency Hopping Spread Spectrum, Direct Sequence Spread Spectrum, Code-Division Multiple Access.	10
4	Circuit Switching and Packet Switching: Switched Communications Networks, Circuit-Switching Network, circuit switching concepts, softswitch architecture, Packet Switching Principles. Routing in Packet-Switched Networks: Routing in Packet-Switching Networks - Characteristics, Routing Strategies. Wireless LANs: 802.11 Architecture and Protocol Stack, frame format, Bluetooth- Bluetooth Architecture. Cellular Wireless Networks: Principles of Cellular Networks	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the fundamental concepts of analog and digital Data Communication and various transmission media.	K2
CO2	Apply different signal encoding and error control techniques in data transmission.	K3
CO3	Explain data link control protocols and concepts of multiplexing and spread spectrum.	K2
CO4	Explain the principles of Networking, Wireless LANs and Cellular Wireless Networks	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	-	-	-	-	-	-	-	-	-	-	2
CO2	2	2	-	2	-	-	-	-	-	-	-	2
CO3	2	-	-	-	-	-	-	-	-	-	-	2
CO4	2	-	-	-	-	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Data and Computer Communication	William Stallings	Pearson Education	8 th Edition 2007

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Data communication and Networking	Behrouz A Forouzan	McGraw Hill	4 th Edition 2007
2	Computer Networks	Andrew S Tanenbaum	Pearson Education	5 th Edition 2011

Video Links (NPTEL, SWAYAM...)	
No.	Link ID
1	https://nptel.ac.in/courses/106105082
2	https://onlinecourses.nptel.ac.in/noc24_ee135/preview
3	https://nptel.ac.in/courses/106105082

SEMESTER S4

FOUNDATIONS OF SECURITY

Course Code	PEITT413	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCITT205: Discrete Mathematical Structures	Course Type	Theory

Course Objectives:

1. To build a robust understanding of essential mathematical concepts and apply these theories to understand and develop traditional and modern cryptographic techniques, including substitution techniques, stream and block ciphers, and secret key ciphers.
2. To solve complex mathematical problems related to number theory and algebraic structures, and apply these solutions to practical cryptographic scenarios.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Divisibility and Modular Arithmetic: Finite Fields – Groups, Rings and Fields. Divisibility - Divisibility and Division Algorithms, Well ordering Principle, Bezout's Identity. Modular Arithmetic- Properties, Euclidean algorithm for the greatest common divisor, Extended Euclidean Algorithm, Least Common multiple, Modular Division.	9
2	Primes and Congruences: Prime Numbers - Prime Numbers and prime-power factorization, Fermat and Mersenne primes, Primality testing and factorization. Congruences - Linear congruences, Chinese Remainder Theorem, 2*2 Linear Systems, Fermat's little theorem, Pseudoprimes and Carmichael numbers.	9
3	Euler's Function & Quadratic Residues: Euler's Function - Euler's Totient function, Applications of Euler's Totient function, Quadratic Residues- Quadratic Congruences, The group of Quadratic residues, Legendre symbol, Jacobi Symbol, Quadratic reciprocity. Sum of Squares and Continued Fractions: Sum of Squares - Sum of two	9

	squares, The Gaussian Integers, Sum of three squares, Sum of four squares. Continued Fractions - Finite continued fractions, Infinite continued fractions.	
4	Traditional and Modern Cryptosystems: Substitution Techniques, Monoalphabetic Cipher, Caesar Cipher, Affine Cipher, Polyalphabetic Ciphers, Autokey Cipher, Playfair Cipher, Hill Cipher, Vigenere Cipher, One Time Pad Cipher, Limitations. Stream and Block Ciphers, Modern Secret Key Ciphers - Substitution Box-Permutation Box-Product Ciphers.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamental concepts of divisibility, modular arithmetic, and finite fields, and apply these concepts to solve mathematical problems.	K2
CO2	Solve problems related to prime numbers, congruences, and primality testing, including the application of the Chinese Remainder Theorem and Fermat's Little Theorem.	K3
CO3	Apply Euler's Totient function, quadratic residues, and concepts of the group of units in various mathematical and cryptographic contexts.	K3
CO4	Solve complex problems involving the sum of squares, continued fractions, and traditional cryptosystems, and assess the effectiveness of various cryptographic algorithms and techniques.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	-	-	-	-	-	-	-	-	-
CO2	3	3	3	3	-	-	-	-	-	-	-	-
CO3	3	3	3	-	2	2	-	-	-	-	-	-
CO4	3	3	3	3	2	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Elementary Number Theory with Applications	Thomas Koshy	Elsevier Science	2 nd Edition 2007
2	Elementary Number Theory	G.A. Jones & J.M. Jones,	Springer UTM	2007
3	A Friendly introduction to Number Theory	Joseph Silverman	Pearson Ed	2009

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	A Course in Number Theory and Cryptography	Neal Koblitz	Springer New York	2012
2	Cryptography and Network Security: Principles and Practice" by	William Stallings	Pearson Education	4 th Edition 2016
3	Introduction to Analytic Number Theory'	Tom M. Apostol	Narosa Publishing House Pvt. Ltd, New Delhi	1996

Video Links (NPTEL, SWAYAM...)	
No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc24_ma89/preview
2	https://onlinecourses.nptel.ac.in/noc22_cs90/preview

SEMESTER S4

COMPUTER GRAPHICS

Course Code	PEITT414	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To build an understanding of the fundamental concepts of computer graphics.
2. To familiarize the working principles of various display technologies.
3. To understand the three dimensional transformations in graphics.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basics of Computer Graphics and its Applications. Raster Scan & Random Scan systems. Display Technologies: Overview of CRT, LCD, Plasma, LED, OLED, AMOLED, E-Paper displays. Output Primitives – Line Drawing Algorithms - DDA, Bresenham's, Midpoint Circle generation algorithm. Filled Area Primitives – Scan Fill, Flood Fill, Boundary Fill. Inside outside tests.	9
2	Two-dimensional transformations - Translation, Rotation, Scaling, Reflection and Shearing, Composite transformations, Matrix representations, and homogeneous coordinates. 2-D Clipping- Window to viewport transformation, Point Clipping, Cohen-Sutherland Line Clipping Algorithm, Sutherland-Hodgeman, and Weiler-Atherton Polygon clipping algorithms.	9
3	Three-Dimensional Concepts: Three-Dimensional Display, Methods Parallel Projection, Perspective Projection, Depth Cueing, Visible Line and Surface, Identification, Surface Rendering, Exploded and Cutaway Views, Three-	9

	Dimensional and Stereoscopic Views Three-Dimensional transformation, Translations, Scaling, Rotation, Viewing Transformation, The Perspective, Algorithms, Three-Dimensional Clipping, Perspective view of Cube	
4	Visible Surface Detection Methods: Depth Buffer, A-Buffer, Scan line, Depth sorting methods. Hidden Surface Removal: Back face detection - Z-buffer method - Painter's algorithm - scan-line algorithm – BSP Trees - Area sub-division method - Ray tracing - Animation - Morphing. Introduction to Digital Image Processing.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe the working principles of graphics devices. Illustrate line drawing, circle drawing and polygon filling algorithms.	K3
CO2	Demonstrate geometric representations and transformations on 2D objects. Explain the working of line and polygon clipping algorithms	K3
CO3	Apply 3D transformation and viewing into the real-world applications	K3
CO4	Describe visible surface detection methods and digital image processing.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3	-	-	-	-	-	-	-	3
CO2	3	3	3	3	3	-	-	-	-	-	-	3
CO3	3	3	3	3	3	-	-	-	-	-	-	3
CO4	3	3	2	2	-	-	-	-	-	-	-	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Computer Graphics with OpenGL	Donald Hearn and M. Pauline Baker	Pearson	4 th Edition 2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Principles of Interactive Computer Graphics	William M. Newman and Robert F. Sproull	McGraw Hill	2 nd Edition 2001
2	Computer Graphics Using OpenGL	Francis Hill Jr.	Pearson Education	3 rd Edition 2007
3	Computer Graphics Programming in OpenGL with C++	V. Scott Gordon, John Clevenger	De Gruyter	2024

Video Links (NPTEL, SWAYAM...)	
No.	Link ID
1	https://youtu.be/V4mP2pQyou0?si=znq_J6ES0MB03Qkm
2	https://youtu.be/8HWvyahgesY?si=1k1hCqfNE19Z00S3
3	https://youtu.be/7iYggzuo6Ks?si=G0Xnf1UA_dRRJaLf
4	https://youtu.be/b1Qpvqm0UAo?si=M6ALob-iLJ9E0i_r

SEMESTER S4

OPERATIONS RESEARCH

Course Code	PEITT415	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GAMAT301: Mathematics for Information Science-3	Course Type	Theory

Course Objectives:

1. This course introduces students to the fundamental concepts and methods of operations research, focusing on applications in Information Technology.
2. Students will learn to model and solve real-world problems using OR techniques and tools.
3. Students will learn to model complex problems in diverse industries and use mathematical optimization and simulation to solve them.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Modelling with Linear Programming: Two-Variable LP Model, Graphical LP Solution, Linear Programming Applications - Investment, Production Planning and Inventory Control, Workforce Planning, Urban Development Planning - The Simplex Method - LP Model in Equation Form, The Simplex Method - Iterative Nature of the Simplex Method, Computational Details of the Simplex Algorithm-Artificial Starting Solution - M-Method, Two-Phase Method.	9
2	Special Cases in the Simplex Method: Degeneracy, Alternative Optima, Unbounded Solution, Infeasible Solution. Sensitivity Analysis - Algebraic Sensitivity Analysis—Changes in the Right-Hand Side, Algebraic Sensitivity Analysis—Objective Function, Sensitivity Analysis with TORA, Solver, and AMPL. Computational Issues in Linear Programming. Dual Problem: Definition of the Dual Problem, Primal–Dual Relationships.	9
3	Transportation Model: Definition of the Transportation Model, Nontraditional Transportation Models, The Transportation Algorithm,	9

	Determination of the Starting Solution, Iterative Computations of the Transportation Algorithm. The Assignment Model - The Hungarian Method	
4	Decision Analysis and Games: Decision Making Under Certainty—Analytic Hierarchy Process (AHP), Decision Making Under Risk, Decision Tree—Based Expected Value Criterion, Variants of the Expected Value Criterion. Decision Under Uncertainty Game Theory - Optimal Solution of Two-Person Zero-Sum Games.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Internal Examination (Written)	Evaluate Level Assessment	Analyze Level Assessment	Total
5	15	10	10	40

Evaluate and Analyze Level Assessment [20 Marks]

Students should evaluate and analyze a real-world optimization problem, assess the proposed solutions, provide a conclusion on which solution is most appropriate for the problem, and implement the chosen solution using Python.

Criteria for evaluation:

1. Problem Definition (K4 - 4 points)

- a. Clearly defines the real-world optimization problem.
- b. Examine and identifies relevant contextual factors (constraints, resources, objectives).

2. Problem Analysis (K4 - 4 points)

- a. Break-down and presents a well-reasoned solution approach.
- b. Compare and justify the proposed solutions with evidence and logical reasoning.

3. Evaluate (K5 - 4 points)

- a. Thoroughly evaluate the proposed solutions.
- b. Compares trade-offs, advantages, and disadvantages.
- c. Considers feasibility, scalability, and practical implications.

4. Python Implementation (K5 - 4 points)

- a. Select the most feasible solution by implementing the proposed solutions.
- b. Successfully translates the chosen solution into Python code.
- c. Demonstrates proficiency in coding practices (readability, efficiency, error handling).

5. Conclusion (K4- 2 points, K5 – 2 points)

- a. Summarizes findings and insights. State which solution is most appropriate for the problem. **(K4)**
- b. Reflects critical thinking and informed decision-making. **(K5)**

Scoring:

1. **Accomplished (4 points):** Exceptional analysis, clear implementation, and depth of understanding.
2. **Competent (3 points):** Solid performance with minor areas for improvement.
3. **Developing (2 points):** Adequate effort but lacks depth or clarity.
4. **Minimal (1 point):** Incomplete or significantly flawed.

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Formulate and solve linear programming problems using both graphical methods and the Simplex Method, applying these techniques to real-world scenarios.	K3
CO2	Analyze and resolve special cases in linear programming, such as degeneracy, alternative optima, unbounded solutions, and infeasible solutions, and perform sensitivity analysis to understand the impact of changes in problem parameters.	K4
CO3	Apply transportation and network modeling techniques, including the transportation algorithm, and the Hungarian method for assignment problems	K3
CO4	Utilize decision analysis tools and game theory to make informed decisions under certainty, risk, and uncertainty, employing methods like the Analytic Hierarchy Process (AHP), decision trees, and strategies for solving two-person zero-sum	K3
CO5	Evaluate any real-world problem and propose a solution using the concepts learned in this course.	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	3	-	-	-	-	-	-	-
CO2	3	3		2	3	-	-	-	-	-	-	-
CO3	3	3	3	2	2	-	-	-	-	-	-	-
CO4	3	3		2	3	2	-	-	-	-	-	-
CO5	3	3	3	3	3	3	-	-	3	3	2	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Operations Research-An Introduction	Hamdy A Thaha	Pearson Education	10 th Edition 2017
2	Operations Research	T Veerarajan	Universities Press	2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Operations Research	Federick S Hiller, Gerald J Lieberman	Tata McGraw Hill	9th Edition 2010
2	Operations Research	Prem Kumar Gupta & D S Hira	S Chand Publishing	7 th Edition 2014
3	Linear and Nonlinear Programming	David G. Luenberger, Yinyu Ye	Springer	5 th Edition 2021

Video Links (NPTEL, SWAYAM...)	
No.	Link ID
1	https://onlinecourses.swayam2.ac.in/cec20_ma10/preview
2	https://onlinecourses.nptel.ac.in/noc22_ma48/preview

SEMESTER S3/S4

ECONOMICS FOR ENGINEERS

Course Code	UCHUT346	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with an understanding of fundamental economic principles essential for effective decision-making in engineering contexts.
2. To enable students to apply economic analysis to production decisions, cost management, and market strategies in engineering practice.
3. To equip students with the ability to evaluate macroeconomic scenarios, financial methods, and investment decisions relevant to engineering projects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic economic problems – Production Possibility Curve – Utility – Law of diminishing marginal utility –Demand: Factors determining demand – Law of Demand – Demand curve- Price elasticity of demand- measurement of price elasticity and its applications – Supply: factors determining supply - Law of supply – Supply curve- Equilibrium price determination- Changes in demand and supply and its effects on equilibrium price and quantity Production: Production function - Law of variable proportion –Returns to scale- Cobb-Douglas Production Function	6
2	Cost: Cost concepts – Private cost and social cost – Sunk cost – Opportunity cost -Explicit and implicit cost –Short run cost curves –Long run average cost curve -Revenue concepts – Break-even point Market: Perfect Competition – Monopoly - Monopolistic Competition (features and equilibrium of a firm) - Oligopoly – Features – Kinked demand model	6

3	National income: Concepts (GDP, GNP and NNP)– Final goods and Intermediate goods - Methods of Estimation –output method – expenditure method-- Difficulties in the measurement of national income. Inflation: Causes and Effects – Measures to Control Inflation - Monetary and Fiscal policies – Repo and reverse repo rate	6
4	Value Analysis and value Engineering: Cost Value, Exchange Value, Use Value, Esteem Value - Aims, Advantages and Application areas of Value Engineering - Value Engineering Procedure Capital Budgeting: Time value of money - Net Present Value Method - Benefit Cost Ratio – Internal Rate of Return – Payback – Accounting Rate of Return.	6

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/Case Study/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
10	15	12.5	12.5	50

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• Minimum 1 and Maximum 2 Questions from each module. • Total of 6 Questions, each carrying 3 marks (6x3 =18 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 8 marks. (4x8 = 32 marks)	50

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of various economic issues using laws and learn the concepts of demand, supply, elasticity and production function.	K2
CO2	Develop decision making capability by applying concepts relating to costs and revenue, and acquire knowledge regarding the functioning of firms in different market situations.	K3
CO3	Outline the macroeconomic principles of monetary and fiscal systems and national income.	K2
CO4	Make use of the possibilities of value analysis and engineering, and take investment decisions through capital budgeting techniques.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	1	-	-	-	-	1	-
CO2	-	-	-	-	-	1	1	-	-	-	1	-
CO3	-	-	-	-	1	-	-	-	-	-	2	-
CO4	-	-	-	-	1	1	-	-	-	-	2	-

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Managerial Economics	Geetika, Piyali Ghosh and Chodhury	Tata McGraw Hill,	2015
2	Engineering Economy	H. G. Thuesen, W. J. Fabrycky	PHI	1966
3	Engineering Economics	R. Paneerselvam	PHI	2012
4	Financial Management	I M Pandey	Vikas Publishing House	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Economy	Leland Blank P.E, Anthony Tarquin P. E.	Mc Graw Hill	7 TH Edition
2	Indian Financial System	Khan M. Y.	Tata McGraw Hill	2011
3	Engineering Economics and analysis	Donald G. Newman, Jerome P. Lavelle	Engg. Press, Texas	2002
4	Contemporary Engineering Economics	Chan S. Park	Prentice Hall of India Ltd	2001
5	Financial Management: Theory and Practice	Prasanna Chandra	Mc Graw Hill	2007

MODEL QUESTION PAPER				
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY				
THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR				
Course Code: UCHUT346				
Course Name: Economics for Engineers				
Max. Marks: 50			Duration: 2 hours 30 minutes	
	PART A			
		Answer all questions. Each question carries 3 marks	CO	Marks
1		What are the central problems of an economy?	CO1	(3)
2		Point out any three applications of price elasticity of demand.	CO1	(3)
3		What is the social cost of production?	CO2	(3)
4		What is repo rate?	CO3	(3)
5		What is esteem value?	CO4	(3)
6		Write a short note on time value of money.	CO4	(3)
PART B				
Answer any one full question from each module. Each question carries 8 marks				
Module 1				
9	a)	Suppose a country is producing at a point inside the production possibility curve. Draw a PPC and examine this situation.	CO1	(5)
	b)	State the law of demand. Point out any two exceptions of this law.	CO1	(3)
10	a)	A consumer purchases 10 units of a commodity when its price is Rs.100. Later when its price falls to Rs.90, he purchases 8 units only. Estimate price elasticity. What type of a commodity is this?	CO1	(5)
	b)	State the law of variable proportion.	CO1	(3)

Module 2				
11	a)	What is oligopoly? Why price is rigid under oligopoly?	CO2	(5)
	b)	The cost function of a firm is given as $TC=1000+10Q-6Q^2+Q^3$. Calculate fixed cost, variable cost and marginal cost when output is 10 units.	CO2	(3)
12	a)	Suppose a firm is earning super normal profit under monopolistic market condition. Explain this situation by drawing a diagram.	CO2	(5)
	b)	Suppose a firm sells its product at a price of Rs.10 per unit and its average variable cost is Rs.6. If the firm spend Ra.10000 as rent and pay Rs. 6000 as interest every month, estimate its break-even level of output.	CO2	(3)
Module 3				
13	a)	What is inflation? How does inflation affect investment and production.	CO3	(5)
	b)	How will you obtain NNP _{fc} from GDP _{mp} .	CO3	(3)
14	a)	From the data given below (In Rs. Crores) estimate GDP _{mp} and national income. Private final consumption expenditure = 1000, Government expenditure = 500, Invest expenditure = 700, Net exports = 300, Depreciation = 200, NFIA=(-200) and Net indirect tax = 100	CO3	(5)
	b)	What is bank rate? Examine the bank rate policy of the government during inflation.	CO3	(3)
Module 4				
15	a)	Examine the procedures of value engineering.	CO4	(5)
	b)	Examine the application areas of value engineering	CO4	(3)

16	a)	1. Suppose the initial investment of a project is Rs. 3000 (Crores) and the cost of capital or the opportunity cost of capital is 10 percent. Calculate NPV of the project based on the cash flows given below. Year 1 2 3 4 5 Cash flow 1000 900 800 700 600 (In Crores)	CO4	(5)
	b)	Point out any three merits of NPV method.	CO4	(3)

SEMESTER S3/S4

ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

Course Code	UCHUT347	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Equip with the knowledge and skills to make ethical decisions and implement gender-sensitive practices in their professional lives.
2. Develop a holistic and comprehensive interdisciplinary approach to understanding engineering ethics principles from a perspective of environment protection and sustainable development.
3. Develop the ability to find strategies for implementing sustainable engineering solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of ethics - Personal vs. professional ethics, Civic Virtue, Respect for others, Profession and Professionalism, Ingenuity, diligence and responsibility, Integrity in design, development, and research domains, Plagiarism, a balanced outlook on law - challenges - case studies, Technology and digital revolution-Data, information, and knowledge, Cybertrust and cybersecurity, Data collection & management, High technologies: connecting people and places-accessibility and social impacts, Managing conflict, Collective bargaining, Confidentiality, Role of confidentiality in moral integrity, Codes of Ethics. Basic concepts in Gender Studies - sex, gender, sexuality, gender spectrum: beyond the binary, gender identity, gender expression, gender stereotypes, Gender disparity and discrimination in education,	6

	employment and everyday life, History of women in Science & Technology, Gendered technologies & innovations, Ethical values and practices in connection with gender - equity, diversity & gender justice, Gender policy and women/transgender empowerment initiatives.	
2	Introduction to Environmental Ethics: Definition, importance and historical development of environmental ethics, key philosophical theories (anthropocentrism, biocentrism, ecocentrism). Sustainable Engineering Principles: Definition and scope, triple bottom line (economic, social and environmental sustainability), life cycle analysis and sustainability metrics. Ecosystems and Biodiversity: Basics of ecosystems and their functions, Importance of biodiversity and its conservation, Human impact on ecosystems and biodiversity loss, An overview of various ecosystems in Kerala/India, and its significance. Landscape and Urban Ecology: Principles of landscape ecology, Urbanization and its environmental impact, Sustainable urban planning and green infrastructure.	6
3	Hydrology and Water Management: Basics of hydrology and water cycle, Water scarcity and pollution issues, Sustainable water management practices, Environmental flow, disruptions and disasters. Zero Waste Concepts and Practices: Definition of zero waste and its principles, Strategies for waste reduction, reuse, reduce and recycling, Case studies of successful zero waste initiatives. Circular Economy and Degrowth: Introduction to the circular economy model, Differences between linear and circular economies, degrowth principles, Strategies for implementing circular economy practices and degrowth principles in engineering. Mobility and Sustainable Transportation: Impacts of transportation on the environment and climate, Basic tenets of a Sustainable Transportation design, Sustainable urban mobility solutions, Integrated mobility systems, E-Mobility, Existing and upcoming models of sustainable mobility solutions.	6
4	Renewable Energy and Sustainable Technologies: Overview of renewable energy sources (solar, wind, hydro, biomass), Sustainable technologies in energy production and consumption, Challenges and opportunities in renewable energy adoption. Climate Change and Engineering Solutions:	6

	Basics of climate change science, Impact of climate change on natural and human systems, Kerala/India and the Climate crisis, Engineering solutions to mitigate, adapt and build resilience to climate change. Environmental Policies and Regulations: Overview of key environmental policies and regulations (national and international), Role of engineers in policy implementation and compliance, Ethical considerations in environmental policy-making. Case Studies and Future Directions: Analysis of real-world case studies, Emerging trends and future directions in environmental ethics and sustainability, Discussion on the role of engineers in promoting a sustainable future.	
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Course Assessment Method **(CIE: 50 marks , ESE: 50)**

Continuous Internal Evaluation Marks (CIE):

Continuous internal evaluation will be based on individual and group activities undertaken throughout the course and the portfolio created documenting their work and learning. The portfolio will include reflections, project reports, case studies, and all other relevant materials.

- The students should be grouped into groups of size 4 to 6 at the beginning of the semester. These groups can be the same ones they have formed in the previous semester.
- Activities are to be distributed between 2 class hours and 3 Self-study hours.
- The portfolio and reflective journal should be carried forward and displayed during the 7th Semester Seminar course as a part of the experience sharing regarding the skills developed through various courses.

Sl. No.	Item	Particulars	Group/Individual (G/I)	Marks
1	Reflective Journal	Weekly entries reflecting on what was learned, personal insights, and how it can be applied to local contexts.	I	5
2	Micro project (Detailed documentation of the project, including methodologies, findings, and reflections)	1 a) Perform an Engineering Ethics Case Study analysis and prepare a report 1 b) Conduct a literature survey on 'Code of Ethics for Engineers' and prepare a sample code of ethics	G	8
		2. Listen to a TED talk on a Gender-related topic, do a literature survey on that topic and make a report citing the relevant papers with a specific analysis of the Kerala context	G	5
		3. Undertake a project study based on the concepts of sustainable development* - Module II, Module III & Module IV	G	12
3	Activities	2. One activity* each from Module II, Module III & Module IV	G	15
4	Final Presentation	A comprehensive presentation summarising the key takeaways from the course, personal reflections, and proposed future actions based on the learnings.	G	5
Total Marks				50

*Can be taken from the given sample activities/projects

Evaluation Criteria:

- **Depth of Analysis:** Quality and depth of reflections and analysis in project reports and case studies.
- **Application of Concepts:** Ability to apply course concepts to real-world problems and local contexts.
- **Creativity:** Innovative approaches and creative solutions proposed in projects and reflections.
- **Presentation Skills:** Clarity, coherence, and professionalism in the final presentation.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop the ability to apply the principles of engineering ethics in their professional life.	K3
CO2	Develop the ability to exercise gender-sensitive practices in their professional lives	K4
CO3	Develop the ability to explore contemporary environmental issues and sustainable practices.	K5
CO4	Develop the ability to analyse the role of engineers in promoting sustainability and climate resilience.	K4
CO5	Develop interest and skills in addressing pertinent environmental and climate-related challenges through a sustainable engineering approach.	K3

Note: *K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1						3	2	3	3	2		2
CO2		1				3	2	3	3	2		2
CO3						3	3	2	3	2		2
CO4		1				3	3	2	3	2		2
CO5						3	3	2	3	2		2

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ethics in Engineering Practice and Research	Caroline Whitbeck	Cambridge University Press & Assessment	2nd edition & August 2011
2	Virtue Ethics and Professional Roles	Justin Oakley	Cambridge University Press & Assessment	November 2006
3	Sustainability Science	Bert J. M. de Vries	Cambridge University Press & Assessment	2nd edition & December 2023
4	Sustainable Engineering Principles and Practice	Bhavik R. Bakshi,	Cambridge University Press & Assessment	2019
5	Engineering Ethics	M Govindarajan, S Natarajan and V S Senthil Kumar	PHI Learning Private Ltd, New Delhi	2012
6	Professional ethics and human values	RS Naagarazan	New age international (P) limited New Delhi	2006.
7	Ethics in Engineering	Mike W Martin and Roland Schinzinger,	Tata McGraw Hill Publishing Company Pvt Ltd, New Delhi	4" edition, 2014

Suggested Activities/Projects:

Module-II

- Write a reflection on a local environmental issue (e.g., plastic waste in Kerala backwaters or oceans) from different ethical perspectives (anthropocentric, biocentric, ecocentric).
- Write a life cycle analysis report of a common product used in Kerala (e.g., a coconut, bamboo or rubber-based product) and present findings on its sustainability.
- Create a sustainability report for a local business, assessing its environmental, social, and economic impacts
- Presentation on biodiversity in a nearby area (e.g., a local park, a wetland, mangroves, college campus etc) and propose conservation strategies to protect it.
- Develop a conservation plan for an endangered species found in Kerala.
- Analyze the green spaces in a local urban area and propose a plan to enhance urban ecology using native plants and sustainable design.
- Create a model of a sustainable urban landscape for a chosen locality in Kerala.

Module-III

- Study a local water body (e.g., a river or lake) for signs of pollution or natural flow disruption and suggest sustainable management and restoration practices.
- Analyse the effectiveness of water management in the college campus and propose improvements - calculate the water footprint, how to reduce the footprint, how to increase supply through rainwater harvesting, and how to decrease the supply-demand ratio
- Implement a zero waste initiative on the college campus for one week and document the challenges and outcomes.
- Develop a waste audit report for the campus. Suggest a plan for a zero-waste approach.
- Create a circular economy model for a common product used in Kerala (e.g., coconut oil, cloth etc).
- Design a product or service based on circular economy and degrowth principles and present a business plan.
- Develop a plan to improve pedestrian and cycling infrastructure in a chosen locality in Kerala

Module-IV

- Evaluate the potential for installing solar panels on the college campus including cost-benefit analysis and feasibility study.
- Analyse the energy consumption patterns of the college campus and propose sustainable alternatives to reduce consumption - What gadgets are being used? How can we reduce demand using energy-saving gadgets?
- Analyse a local infrastructure project for its climate resilience and suggest improvements.
- Analyse a specific environmental regulation in India (e.g., Coastal Regulation Zone) and its impact on local communities and ecosystems.
- Research and present a case study of a successful sustainable engineering project in Kerala/India (e.g., sustainable building design, water management project, infrastructure project).
- Research and present a case study of an unsustainable engineering project in Kerala/India highlighting design and implementation faults and possible corrections/alternatives (e.g., a housing complex with water logging, a water management project causing frequent floods, infrastructure project that affects surrounding landscapes or ecosystems).

SEMESTER S4

COMPUTER NETWORKS LAB

Course Code	PCITL407	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. To familiarize students with internetworking concepts and provide hands-on experience in identifying, configuring, and troubleshooting key network components such as routers, switches, and network interfaces
2. To enable students to configure, implement, and verify both static and dynamic routing protocols, ensuring effective and efficient routing within a network.
3. To equip students with the skills to design, implement, and manage Virtual LANs (VLANs) as well as configure and apply standard and extended access control lists (ACLs) for network security and traffic management.
4. To teach students how to use remote login, file transfer, and automatic network configuration protocols such as Telnet, FTP, and DHCP, enhancing their ability to manage and automate network operations.

Experiment No.	Experiments
1	Familiarization of network components- Hub, Switch, Bridge, Router, Access Point, Network cables, Colour coding, Crimping.
2	Familiarization of Internetworking operating system – access the CLI, switch between different command modes, basic IOS commands - to display device information and configurations, to configure interface and basic troubleshooting.
3	Study of IPv4, IPv6 addresses and subnetting.
4	Configure and verify IPv4 static routing in a network that includes subnetting.
5	Configure and verify Routing Information Protocol (RIP) for dynamic routing in a network that includes subnetting.
6	Configure and verify Enhanced Interior Gateway Routing Protocol (EIGRP) routing in a network that includes subnetting.
7	Configure and verify Open Shortest Path First (OSPF) routing in a network that includes subnetting.

8	Implement Unicast IPv6 Addresses on routers and verify static routing, RIPNG and OSPFv3.
9	Create and apply Standard and Extended ACLs on routers to filter traffic based on IP addresses and protocols.
10	Configure VLANs and implement Inter-VLAN routing using router on a stick technique.
11	Configure TELNET, login to a remote machine and view the files on the remote machine. Configure FTP and transfer files between two machines.
12	Configure and verify the operation of a DHCP (Dynamic Host Configuration Protocol) server in a network.
13	Configure and verify Network Address Translation (NAT) on a router.
14	Familiarize with any popular network simulator.

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- **Submission of Record:** Students shall be allowed for the end semester examination only upon submitting the duly certified record.
- **Endorsement by External Examiner:** The external examiner shall endorse the record

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Demonstrate internetworking and network components	K3
CO2	Configure and verify both static and dynamic routing protocols	K3
CO3	Develop the ability to implement and manage VLANs, standard and extended access lists.	K3
CO4	Use remote login, file transfer and automatic network configuration protocols	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	3	2	-	-	-	-	-	2	-	2
CO2	3	2	3	2	-	-	-	-	3	2	-	2
CO3	3	2	3	2	-	-	-	-	3	2	-	2
CO4	3	2	3	2	-	-	-	-	3	2	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	CCNA 200-301 Official Cert Guide, Volume 1	Wendell Odom	Cisco Press	2 nd Edition 2024
2	CCNA –Cisco Certified Network Associate. Study Guide	Todd Lammle	Wiley India	6 th Edition 2007

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Computer Networking: The Complete Beginner's Guide to Learning the Basics of Network Security, Computer Architecture, Wireless Technology and Communications Systems	Benjamin Walker	Science & Technology	2019
2	Computer Networking Bible: The Complete Crash Course to Effectively Design, Implement and Manage Networks.	Rick C. Worley	Independently published	2023

Video Links (NPTEL, SWAYAM...)	
Link No.	Link ID
1	https://archive.nptel.ac.in/courses/106/106/106106091/
2	https://archive.nptel.ac.in/courses/106/105/106105081/
3	https://archive.nptel.ac.in/courses/106/105/106105183/
4	https://archive.nptel.ac.in/courses/106/106/106106243/

Continuous Assessment (25 Marks):

5. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

6. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

7. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness, and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

8. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results, and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks):

6. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.

- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

7. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

8. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

9. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related question.
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

10. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S4

OPERATING SYSTEMS LAB

Course Code	PCITL408	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GXEST204: Programming in C	Course Type	Lab

Course Objectives:

1. Students will be able to perform basic and advanced UNIX commands, system calls, threads and inter process communication essential for operating system functionalities.
2. Students will be able to solve classical synchronization problems using appropriate techniques such as semaphores, threads, and shared memory.
3. Students will be able to understand and implement various memory management schemes, page replacement algorithms, disk scheduling algorithms, and file organization techniques.

Experiment No.	Experiments
1	Familiarization of system calls (fork, exec, getpid, exit, wait, close, stat etc) in operating system and Demonstration of creating partition and installing Operating System (Not for Examination purpose)
2	Implementation of Process and thread (Life cycle of process): (i) Process creation and Termination; (ii) Thread creation and Termination
3	Implement process scheduling algorithms (FCFS, SJF, Round Robin, Priority) and compute average waiting time and average turn-around time.
4	Inter-process communication using pipes, message queues and shared memory.
5	Implement Producer-Consumer Problem and solve the problem using Semaphores
6	Implementation of Dining Philosophers' Problem using threads and semaphores
7	Implementation of banker's algorithm
8	Implement memory management schemes (first fit, best fit and worst fit)
9	Implement page replacement algorithms: FIFO, LRU and Optimal

10	Implementation of Disk Scheduling using FCFS, SCAN, C-SCAN, LOOK, C-LOOK and SSTF algorithm
11	Implement File Organization Technique (Single Level and Multi-Level Directory) and Allocation Strategies (Sequential, Linked and Indexed)
12	Familiarization of Cloud Platforms and Virtualization tools(eg:-VMWare)

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- **Submission of Record:** Students shall be allowed for the end semester examination only upon submitting the duly certified record.
- **Endorsement by External Examiner:** The external examiner shall endorse the record

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Implement process and thread lifecycle management, including creation, termination, and scheduling.	K3
CO2	Implement inter-process communication techniques such as pipes, message queues, and shared memory.	K3
CO3	Solve classical synchronization problems like Producer-Consumer and Dining Philosophers using semaphores and threads.	K3
CO4	Implement memory management and disk scheduling algorithms, and understand file organization and allocation strategies.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	3	2	-	-	-	-	3	3	-	2
CO2	3	-	3	2	-	-	-	-	3	3	-	2
CO3	3	2	3	2	-	-	-	-	3	3	-	2
CO4	3	2	3	2	-	-	-	-	3	3	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Operating System Concepts	Abraham Silberschatz, Peter B Galvin, Greg Gagne	John Wiley & Sons	10 th Edition 2021
2	Programming in C	E. Balagurusamy	Mc Graw Hill	8 th Edition 2019

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Modern Operating Systems	Andrew S Tanenbaum	Pearson	4 th Edition 2016
2	C Primer Plus	Stephen Prata	Pearson	6 th Edition 2020

Video Links (NPTEL, SWAYAM...)	
Link No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc24_cs80/preview
2	https://onlinecourses.nptel.ac.in/noc24_cs123/preview

Continuous Assessment (25 Marks):

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks):

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.

- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related question.
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 5

INFORMATION TECHNOLOGY